Package 'RFDose'

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Type Package

Title Deterministic radiofrequency electromagnetic field (RF-EMF) dose calculator

Version 0.4.0

Description Package for calculating RF-EMF doses for cohorts.

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yaml

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rmarkdown,
testthat (>= 3.0.0),
tibble

VignetteBuilder knitr

Depends R (>= 3.5)

Config/testthat/edition 3

Contents

`act_pwr_props` *Get low/lowmed/medhigh/high activity power proportions*

Description

Converts activity durations to proportions.

Usage

```
act_pwr_props(low_dur, lowmed_dur, medhigh_dur, high_dur)
```

Arguments

`duration_low`	Duration (in seconds per day) of low output power activities
`duration_lowmed`	Duration (in seconds per day) of low-medium output power activities
`duration_medhigh`	Duration (in seconds per day) of medium-high output power activities
`duration_high`	Duration (in seconds per day) of high output power activities

Value

Named list with the use proportion of each activity.

`calculate_emf_doses` *Calculate Total RF-EMF Doses for ALL samples*

Description

Wrapper function that calculates RF-EMF doses for a data frame of observations. This function will be removed in the future.

Usage

```
calculate_emf_doses(data, tissue, params = NULL, default_value_file = NULL)
```

Arguments

`data`	A data frame. Must have the exact same columns as the provided example dataset. Columns must not be missing. Missing values will be replaced with default values.
`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`params`	(experimental) Path to a parameter file in .yaml format. Must match the structure of the internal parameter file. Optional; if not specified, calculations use default parameters.
`default_value_file`	List of default values to replace missing values (optional).

Details

Additional details will follow.

Value

A data frame. Columns contain the calculated RF-EMF dose of each row in mJ/kg/day, for each exposure source and each tissue.

`check_boolean_not_na` *Check if value is logical/Boolean and non-NA*

Description

Check if value is logical/Boolean and non-NA

Usage

```
check_boolean_not_na(x)
```

Arguments

`x`	Single value to be checked: should be a logical/Boolean, atomic, and not NA to pass check

Value

Returns invisible TRUE if check passes. Side effect of stopping calculation if check fails.

`check_character_not_na` *Check if value is character and non-NA*

Description

Check if value is character and non-NA

Usage

```
check_character_not_na(x)
```

Arguments

`x`	Single value to be checked: should be character, atomic, and not NA to pass check

Value

Returns invisible TRUE if check passes. Side effect of stopping calculation if check fails.

`check_country` *Check if country is valid*

Description

Checks if country code matches a country for which measurement data is available. Use "Other" for any other country.

Usage

```
check_country(country)
```

Arguments

`country`	Country to check. Must be one of these: "AT" (Austria), "BE" (Belgium), "FR" (France), "HU" (Hungary), "IT" (Italy), "NL" (Netherlands), "PL" (Poland), "ES" (Spain), "CH" (Switzerland), "UK" (United Kingdom), or "Other" (for any other country; calculations use averaged values from countries with available measurements)

Value

Returns invisible TRUE if all checks pass. Side effect of stopping calculations if any check fails.

`check_duration` *Check input duration(s)*

Description

Checks supplied input duration or vector of durations; needed to stop calculation or raise warning in case of irregular input values.

Usage

```
check_duration(duration)
```

Arguments

`duration`	single duration (numeric) or vector of durations

Value

Returns invisible TRUE if all checks pass. Side effects of stopping or showing warning message if any check fails

check_headp_num *Check if number of headphones (earbuds) is valid*

Description

Checks if the number of headphones (earbuds) worn during mobile phone calls is either 0, 1, or 2.

Usage

```
check_headp_num(headp_num)
```

Arguments

`headp_num`	Number of headphones (earbuds) worn during mobile phone call.

Value

Returns invisible TRUE if all checks pass. Side effect of stopping calculations if any check fails.

check_numeric_not_na *Check if value is numeric and non-NA*

Description

Check if value is numeric and non-NA

Usage

```
check_numeric_not_na(x)
```

Arguments

`x`	Single value to be checked: should be numeric, atomic, and not NA to pass check

Value

Returns invisible TRUE if check passes. Side effect of stopping calculation if check fails.

`check_proportions` *Check input proportion(s)*

Description

Checks supplied input proportion or vector of proportions; needed to stop calculation or raise warning in case of irregular input values.

Usage

```
check_proportions(proportions)
```

Arguments

`proportions`	Single proportion value (between 0 and 1) or vector with multiple proportions to check

Value

Returns invisible TRUE if all checks pass. Side effects of stopping or showing warning message if any check fails.

`check_tissue` *Check if tissue is valid*

Description

Check if tissue with this name can be found in YAML file, and if it has any nested parameters. Parameters themselves are NOT checked for validity.

Usage

```
check_tissue(tissue, device, params)
```

Arguments

`tissue`	Name of tissue to be checked (character, length 1)
`device`	Device for which to check tissue (character, length 1)
`params`	Parameter list, as returned by `load_params()`

Value

Returns invisible TRUE if all checks pass. Side effect of stopping calculations if any check fails.

`check_urbanicity` *Check if input urbanicity is valid*

Description

Checks if a supplied urbanicity value is valid (must be "urban", "rural", or "suburban").

Usage

```
check_urbanicity(urbanicity)
```

Arguments

`urbanicity`	Single urbanicity value (must be one of these: "urban", "rural", "suburban")

Value

Returns invisible TRUE if checks pass. Side effect of stopping calculation if any check fails.

`cordless_dose` *Calculate Dose from Cordless Calls*

Description

Calculates RF-EMF dose from making calls with cordless phone (also known as DECT phone).

Usage

```
cordless_dose(tissue, duration, ear_prop, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`duration`	Duration of cordless calls in s/day.
`ear_prop`	Proportion of time cordless phone is held against ear during calls.
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The dose is calculated as

$$Dose_{DECT} = mSAR_{DECT} * duration_{DECT}$$

Where:

- $Dose_{DECT}$ is the dose from DECT/cordless phone calls
- $mSAR_{DECT}$ is the momentary SAR value in mJ/kg
- $duration_{DECT}$ is the duration of the calls in s/day

Value

Tissue-specific RF-EMF dose from DECT/cordless phone calls in mJ/kg/day

See Also

dect_msar()

Examples

```
cordless_dose(
  tissue      = "brain",
  duration    = 600,
  ear_prop    = 0.9)
```

---

dect_msar *Calculate mSAR from Cordless Calls*

Description

Calculates mSAR (momentary SAR) from making calls with cordless phone (also known as DECT phone).

Usage

```
dect_msar(tissue, ear_prop, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
ear_prop	Proportion of time cordless phone is held against ear during calls.
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as

$$mSAR_{DECT} = nSAR_{DECT} * outputpower_{DECT}$$

Where:

- $mSAR_{DECT}$ is the mSAR from DECT/cordless phone calls
- $nSAR_{DECT}$ is the nSAR (normalized specific absorption rate) in W/kg/W
- $outputpower_{DECT}$ is the duration of the calls in mW

Value

Tissue-specific mSAR from DECT/cordless phone calls in mW/kg

See Also

dect_pwr(), dect_sar()

Examples

```
dect_msar(
  tissue      = "brain",
  ear_prop    = 0.9)
```

`dect_pwr` *Calculate Output Power of DECT/Cordless Phone*

Description

Calculates the output power of DECT/cordless phone during a call.

Usage

```
dect_pwr(params = load_params())
```

Arguments

`params`	Parameter list (optional). If not specified, calculations use default parameters.

Value

DECT/cordless phone output power in mW

`dect_sar` *Calculate nSAR from Cordless Calls*

Description

Calculates nSAR (normalized specific absorption rate) from making calls with cordless phone (also known as DECT phone).

Usage

```
dect_sar(tissue, ear_prop, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate the nSAR (default: "brain" or "body")
`ear_prop`	Proportion of time cordless phone is held against ear during calls.
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Value

Tissue-specific nSAR from DECT/cordless phone calls in W/kg/W

Examples

```
dect_sar(
  tissue      = "brain",
  ear_prop    = 0.9)
```

example_data *Example data for testing*

Description

A dataset containing example user data (made up)

Usage

```
example_data
```

Format

A data frame

sample_id test
use_5g test
travel_time test
urbanicity test
country test
headp_ear_num test
mpc_duration test
mpc_ear_prop test
mpc_headp_prop test
mpd_wifi_prop_home test
mpd_wifi_prop_travel test
mpd_wifi_prop_work test
mpd_dur_low test
mpd_dur_lowtomed test
mpd_dur_medtohigh test
mpd_dur_high test
dect_duration test
dect_ear_prop test
lptp_dur_low test
lptp_dur_lowtomed test
lptp_dur_medtohigh test
lptp_dur_high test
tblt_dur_low test
tblt_dur_lowtomed test
tblt_dur_medtohigh test
tblt_dur_high test
hotspot_duration test
smartwatch_duration test
tracker_duration test
vr_duration test
headphone_duration test
gaming_duration test

Source

Generated for package testing.

`farfield_dose` *Calculate RF-EMF Dose from Far-field Sources*

Description

Calculates the tissue-specific RF-EMF dose from far-field sources

Usage

```
farfield_dose(tissue, country, urbanicity, travel_time, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`country`	Country of residence (Austria = "AT", Belgium = "BE", France = FR, Hungary = "HU", Italy = "IT", Netherlands = "NL", Poland = "PL", Spain = "ES", Switzerland = "CH", United Kingdom = "UK", elsewhere = "Other")
`urbanicity`	Urbanicity of home / workplace (rural, suburban, or urban)
`travel_time`	Time spent commuting in seconds per day
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from far-field sources is calculated as:

$$Dose_{farfield} = mSAR_{farfield} * 86400$$

Where:

- $Dose_{farfield}$ is the dose from far-field sources
- $mSAR_{farfield}$ is the momentary SAR value in mJ/kg

Value

Tissue-specific RF-EMF dose from far-field sources in mJ/kg/day

See Also

`farfield_msar()`

Examples

```
farfield_dose(
  tissue      = "body",
  country     = "CH",
  urbanicity  = "suburban",
  travel_time = 1800)
```

farfield_msar *Calculate mSAR of Far-field Sources*

Description

Calculates the mSAR of far-field sources

Usage

```
farfield_msar(tissue, country, urbanicity, travel_time, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
country	Country of residence (Austria = "AT", Belgium = "BE", France = FR, Hungary = "HU", Italy = "IT", Netherlands = "NL", Poland = "PL", Spain = "ES", Switzerland = "CH", United Kingdom = "UK", elsewhere = "Other")
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the country, urbanicity, and time spent commuting.

Value

mSAR in W/kg/W/m**2

See Also

farfield_pwr(), farfield_sar()

Examples

```
farfield_msar(
  tissue      = "body",
  country     = "CH",
  urbanicity  = "suburban",
  travel_time = 1800)
```

`farfield_pwr` *Calculate Output Power of Far-field Sources*

Description

Calculates the output power of far-field sources

Usage

```
farfield_pwr(country, urbanicity, travel_time, params = load_params())
```

Arguments

`country`	Country of residence (Austria = "AT", Belgium = "BE", France = FR, Hungary = "HU", Italy = "IT", Netherlands = "NL", Poland = "PL", Spain = "ES", Switzerland = "CH", United Kingdom = "UK", elsewhere = "Other")
`urbanicity`	Urbanicity of home / workplace (rural, suburban, or urban)
`travel_time`	Time spent commuting in seconds per day
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the country, urbanicity, and time spent commuting.

Value

Output power in mW/m**2

Examples

```
farfield_pwr(
country     = "CH",
urbanicity  = "suburban",
travel_time = 1800)
```

`farfield_sar` *Calculate nSAR from Far-Field Sources*

Description

Calculates tissue-specific nSAR from far-field sources.

Usage

```
farfield_sar(tissue, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate nSAR (default: "brain" or "body")
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue.

Value

nSAR in W/kg/W/m**2

Examples

```
farfield_sar(
tissue      = "brain")
```

---

fill_missing_variables

Fill missing values in input data frame with missing values

Description

Replaces missing values in input data frame with values specified as default values. In case a column is completely missing, it is added to the data frame, with all values being the default value.

Usage

```
fill_missing_variables(data, defaults, warn_threshold = 0.1)
```

Arguments

`data`	Input data frame
`defaults`	Named list (key = column name, value = default value to replace missing values in this column)
`warn_threshold`	threshold proportion of missing data per column to raise warning

Value

data frame with missing data replaced with default values. Side effect of raising warning if proportion of missing data exceeds warn_threshold for at least one column.

`gaming_dose` *Calculate RF-EMF Dose from online gaming (console)*

Description

Calculates the tissue-specific RF-EMF dose from online gaming

Usage

```
gaming_dose(tissue, duration_gaming, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`duration_gaming`	duration of gaming with portable console in s/day
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from online gaming is calculated as:

$$Dose_{gaming} = mSAR_{gaming} * duration_{gaming}$$

Where:

- $mSAR_{gaming}$ is the mSAR of gaming
- $Dduration_{gaming}$ is the duration of gaming with portable console in s/day

Value

Tissue-specific RF-EMF dose from gaming in mJ/kg/day

Examples

```
gaming_dose(
  tissue          = "body",
  duration_gaming = 600)
```

headphones_dose *Calculate RF-EMF Dose from Bluetooth Headphones*

Description

Calculates the tissue-specific RF-EMF dose from bluetooth headphones (calls excluded)

Usage

```
headphones_dose(tissue, duration_headphones, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`params`	Parameter list (optional). If not specified, calculations use default parameters.
`duration_smartwatch`	duration of time Bluetooth headphones were used in s/day

Details

The RF-EMF dose from Bluetooth headphone use is calculated as:

$$Dose_{headphones} = Dose_{earset} + Dose_{phone}$$

Where:

- $Dose_{earset}$ is the dose contribution from the headphones themselves
- $Dose_{phone}$ is the dose contribution from the phone connection to the headphones

Value

Tissue-specific RF-EMF dose from Bluetooth headphone use in mJ/kg/day

See Also

headphones_earset_dose(), headphones_phone_dose()

Examples

```
headphones_dose(
  tissue              = "body",
  duration_headphones = 6000)
```

---

headphones_earset_dose
Calculate RF-EMF Dose from Smart Watch (earset contribution only)

Description

Calculates the tissue-specific RF-EMF dose from Bluetooth headphone use (earset contribution only).

Usage

```
headphones_earset_dose(tissue, duration_headphones, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`params`	Parameter list (optional). If not specified, calculations use default parameters.
`duration_smartwatch`	duration of time Bluetooth headphones were worn in s/day

Details

The RF-EMF dose from Bluetooth headphone use (earset contribution) is calculated as:

$$Dose_{earset} = mSAR_{earset} * duration_{earset}$$

Where:

- $mSAR_{earset}$ is the momentary SAR for the headphone connection
- $duration_{earset}$ is the use duration in s/day

Value

Tissue-specific RF-EMF dose from Bluetooth headphone use (earset contribution only) in mJ/kg/day

Examples

```
headphones_earset_dose(
  tissue              = "body",
  duration_headphones = 6000)
```

headphones_phone_dose *Calculate RF-EMF Dose from Smart Watch (phone contribution only)*

Description

Calculates the tissue-specific RF-EMF dose from Bluetooth headphone use (phone contribution only).

Usage

```
headphones_phone_dose(tissue, duration_headphones, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`params`	Parameter list (optional). If not specified, calculations use default parameters.
`duration_smartwatch`	duration of time Bluetooth headphones were worn in s/day

Details

The RF-EMF dose from Bluetooth headphone use (phone contribution) is calculated as:

$$Dose_{phone} = mSAR_{phone} * duration_{phone}$$

Where:

- $mSAR_{phone}$ is the momentary SAR for the headphone connection
- $duration_{phone}$ is the use duration in s/day

Value

Tissue-specific RF-EMF dose from Bluetooth headphone use (phone contribution only) in mJ/kg/day

Examples

```
headphones_phone_dose(
  tissue              = "body",
  duration_headphones = 6000)
```

`hotspot_dose` *Calculate RF-EMF Dose from Mobile Phone Hotspot*

Description

Calculates the tissue-specific RF-EMF dose from hotspot

Usage

```
hotspot_dose(tissue, duration_hotspot, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`duration_hotspot`	duration of time hotspot was active in s/day
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from hotspot use is calculated as:

$$Dose_{hotspot} = mSAR_{hotspot} * duration_{hotspot}$$

Where:

- $mSAR_{hotspot}$ is the mSAR of hotspot use
- $Dduration_{hotspot}$ is the duration of hotspot use in s/day

Value

Tissue-specific RF-EMF dose from hotspot use in mJ/kg/day

Examples

```
hotspot_dose(
  tissue           = "body",
  duration_hotspot = 600)
```

laptop_dose *Calculate RF-EMF Dose from Laptop Use*

Description

Calculates the tissue-specific RF-EMF dose from laptop use.

Usage

```
laptop_dose(
  tissue,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
dur_low	Duration (in seconds per day) of low output power activities on laptop
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on laptop
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on laptop
dur_high	Duration (in seconds per day) of high output power activities on laptop
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The laptop RF-EMF dose is calculated as:

$$Dose_{laptop} = mSAR_{laptop} * duration_{laptop}$$

Where:

- $Dose_{laptop}$ is the dose from laptop use
- $mSAR_{laptop}$ is the momentary SAR value in mJ/kg
- $duration_{laptop}$ is the duration of laptop use

We distinguish between 4 types of activities:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific RF-EMF dose from laptop use in mJ/kg/day

See Also

laptop_msar()

Examples

```
laptop_dose(
  tissue      = "brain",
  dur_low     = 0,
  dur_lowmed  = 1230,
  dur_medhigh = 0,
  dur_high    = 0)
```

---

laptop_msar *Calculate mSAR from Laptop Use*

Description

Calculates the tissue-specific momentary SAR (mSAR) from laptop use.

Usage

```
laptop_msar(
  tissue,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
dur_low	Duration (in seconds per day) of low output power activities on laptop
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on laptop
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on laptop
dur_high	Duration (in seconds per day) of high output power activities on laptop
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as:

$$mSAR_{laptop} = nSAR_{laptop} * outputpower_{laptop}$$

Where:

- $mSAR_{laptop}$ is the mSAR from laptop use
- $nSAR_{laptop}$ is the normalised SAR value in mJ/kg
- $outputpower_{laptop}$ is the duration of laptop use

We distinguish between 4 types of activities:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific mSAR from laptop use in mW/kg

See Also

`laptop_pwr()`, `laptop_sar()`

Examples

```
laptop_msar(
  tissue      = "brain",
  dur_low     = 0,
  dur_lowmed  = 1230,
  dur_medhigh = 0,
  dur_high    = 0)
```

`laptop_pwr` *Calculate Laptop Output Power*

Description

Calculates the laptop output power during use

Usage

```
laptop_pwr(
  band,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

band	WiFi frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
dur_low	Duration (in seconds per day) of low output power activities on laptop
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on laptop
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on laptop
dur_high	Duration (in seconds per day) of high output power activities on laptop
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the frequency band (2.4 GHz or 5.0 GHz) and the type of activity.

Value

Output power in mW

Examples

```
laptop_pwr(
  band        = "2",
  dur_low     = 0,
  dur_lowmed  = 1230,
  dur_medhigh = 0,
  dur_high    = 0)
```

---

laptop_sar *Calculate nSAR during Laptop Use*

Description

Calculates tissue-specific nSAR from laptop use.

Usage

```
laptop_sar(tissue, band, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate nSAR (default: "brain" or "body")
`band`	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the frequency band.

Value

nSAR in W/kg/W

Examples

```
laptop_sar(
tissue      = "brain",
band        = "2")
```

`load_params` *Load dose calculation parameters*

Description

Loads default parameters or parameters from external source and returns them as a nested list.

Usage

```
load_params(path = NULL)
```

Arguments

`path`	Path of parameter file to load (must be a YAML file). If null, default parameters are loaded.

Value

List of parameters loaded from yaml input file

See Also

`load_tissue_params()`

`load_tissue_params` *Load tissue-specific parameters*

Description

Loads device- and tissue-specific parameters (SAR values) and returns them as a list.

Usage

```
load_tissue_params(params, device_type, tissue_name)
```

Arguments

`params`	List of parameters (default parameter file or custom YAML that matches default's structure)
`device_type`	Name of device (default:)
`tissue_name`	Name of tissue (default: "brain" or "body")

Value

List of parameters (SAR values) specific to selected device type and tissue.

See Also

`load_params()`

`location_props` *Calculate proportion of time spent at home vs work vs outdoors vs travelling*

Description

Based on travel time (input variable by user). Assumes a fixed ratio of time spent at home vs at work vs outdoors.

Usage

```
location_props(travel_time, home_prop, work_prop, outd_prop)
```

Arguments

`travel_time`	daily time spent travelling / commuting in seconds (s)
`home_prop`	proportion of time spent at home WITHOUT considering commute
`work_prop`	proportion of time spent at work WITHOUT considering commute
`outd_prop`	proportion of time spent outside WITHOUT considering commute

Details

We assume that the time spent commuting takes away from time spent at home, but not the time spent at work or outdoors.

Value

List with following elements: "travel", "home", "work", "out", corresponding value is the proportion of time spent in each location.

---

mobilecall_dose *Calculation of RF-EMF Dose from Mobile Phone Calls*

Description

Includes RF-EMF exposure from phone itself, for Bluetooth connection to headphones, and from headphones themselves

Usage

```
mobilecall_dose(
  tissue,
  duration,
  ear_prop,
  headp_prop,
  urbanicity,
  use_5g,
  travel_time,
  headp_ear_num,
  wifi_prop_home,
  wifi_prop_work,
  wifi_prop_travel,
  params = load_params()
)
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`duration`	Duration of mobile phone call in seconds per day
`ear_prop`	Proportion of call performed with phone held against ear
`headp_prop`	Proportion of call performed with active Bluetooth connection to headphones
`urbanicity`	Urbanicity of home / workplace (rural, suburban, or urban)
`use_5g`	TRUE if 5g services are used for calls, FALSE if not
`travel_time`	Time spent commuting in seconds per day
`headp_ear_num`	Number of Bluetooth headphones used during call
`wifi_prop_home`	Proportion of time connected to WiFi (vs mobile data) at home
`wifi_prop_work`	Proportion of time connected to WiFi (vs mobile data) at school / work
`wifi_prop_travel`	Proportion of time connected to WiFi (vs mobile data) while commuting
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mobile call RF-EMF dose is calculated as:

$$Dose_{mobilecall} = Dose_{phone} + Dose_{bluetooth}$$

Where:

- $Dose_{mobilecall}$ is the total dose from all mobile call activities
- $Dose_{phone}$ is the dose from the mobile phone (no Bluetooth)
- $Dose_{bluetooth}$ is the dose from using Bluetooth headphones during the call.

For each source (phone and bluetooth), the dose is calculated as:

$$mSAR * duration$$

Where:

- $mSAR$ is the momentary SAR value in mJ/kg
- $duration$ is the call duration in seconds

Value

Tissue-specific RF-EMF dose in mJ/kg/day

See Also

mpc_msar()

Examples

```
mobilecall_dose(
tissue           = "brain",
duration         = 400,
ear_prop         = 0.67,
headp_prop       = 0.17,
urbanicity       = "suburban",
use_5g           = TRUE,
travel_time      = 1800,
headp_ear_num    = 2,
wifi_prop_home   = 1,
wifi_prop_work   = 0.8,
wifi_prop_travel = 0)
```

mobiledata_dose — *Calculation of RF-EMF Dose from Mobile Phone Data Use*

Description

The mobile data RF-EMF dose is calculated as:

$$Dose_{data} = mSAR_{data} * duration_{data}$$

Usage

```
mobiledata_dose(
  tissue,
  duration_low,
  duration_lowmed,
  duration_medhigh,
  duration_high,
  use_5g,
  wifi_prop_home,
  wifi_prop_work,
  wifi_prop_travel,
  urbanicity,
  travel_time,
  params = load_params()
)
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`duration_low`	Duration (in seconds per day) of low output power activities
`duration_lowmed`	Duration (in seconds per day) of low-medium output power activities
`duration_medhigh`	Duration (in seconds per day) of medium-high output power activities
`duration_high`	Duration (in seconds per day) of high output power activities
`use_5g`	TRUE if 5G services are used for data transfer, FALSE if not
`wifi_prop_home`	Proportion of time connected to WiFi (vs mobile data) at home
`wifi_prop_work`	Proportion of time connected to WiFi (vs mobile data) at school / work
`wifi_prop_travel`	Proportion of time connected to WiFi (vs mobile data) while commuting
`urbanicity`	Urbanicity of home / workplace (rural, suburban, or urban)
`travel_time`	Time spent commuting in seconds per day
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

Where:

- $Dose_{data}$ is the dose from mobile data use
- $mSAR_{data}$ is the momentary SAR value in mJ/kg
- $duration_{data}$ is the duration of mobile data use

We distinguish between 4 types of mobile data use:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific RF-EMF dose from data use in mJ/kg/day

See Also

mpd_msar()

Examples

```
mobiledata_dose(
tissue           = "brain",
duration_low     = 4860,
duration_lowmed  = 540,
duration_medhigh = 4860,
duration_high    = 540,
use_5g           = FALSE,
wifi_prop_home   = 0.5,
wifi_prop_work   = 0.5,
wifi_prop_travel = 0.5,
urbanicity       = "suburban",
travel_time      = 1800)
```

---

mpc_bt_earset_msar — *Calculate mobile call mSAR for Bluetooth headphones (headphones only)*

Description

Calculates the mSAR for mobile calling using Bluetooth headphones (headphone/earset only; the contribution of the mobile phone is calculated separately).

Usage

```
mpc_bt_earset_msar(tissue, headp_ear_num, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate mSAR (default: "brain" or "body")
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Value

mSAR from Bluetooth calls (headphones/earset only) in mJ/kg/day)

See Also

`mpc_bt_phone_sar()`

Examples

```
mpc_bt_earset_msar(
  tissue = "brain"
)
```

`mpc_bt_phone_msar` *Calculate mobile call dose from Bluetooth headphones (phone only)*

Description

Calculates the RF-EMF dose from mobile calling using Bluetooth headphones (phone contribution only; the contribution of the headphones/earset is calculated separately).

Usage

```
mpc_bt_phone_msar(tissue, headp_ear_num, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate mSAR (default: "brain" or "body")
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Value

Dose from Bluetooth calls (phone contribution only) in mW/kg)

See Also

`mpc_bt_earset_msar()`

Examples

```
mpc_bt_phone_msar(
tissue = "brain"
)
```

mpc_msar *Calculation of mSAR from Mobile Phone Calls*

Description

Calculates mSAR (momentary specifc absorption rate) from mobile calls for a specific tissue.

Usage

```
mpc_msar(
  tissue,
  prop_native,
  prop_data,
  prop_wifi,
  headp_prop,
  ear_prop,
  speaker_prop,
  urbanicity,
  use_5g,
  travel_time,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
prop_native	Proportion of mobile call using native connection
prop_data	Proportion of mobile call using mobile data connection
prop_wifi	Proportion of mobile call using WiFi connection
headp_prop	Proportion of call performed with active Bluetooth connection to headphones
ear_prop	Proportion of call performed with phone held against ear
speaker_prop	Proportion of call performed in speaker mode
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
use_5g	TRUE if 5g services are used for calls, FALSE if not
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as

$$mSAR = nSAR_{native} \times outputpower_{native} + nSAR_{data} \times outputpower_{data} + nSAR_{wifi} \times outputpower_{wifi}$$

where

- $nSAR_{native}$, $nSAR_{data}$, $nSAR_{wifi}$ are the nSAR (normalized specific absorption rates, in W/kg/W) from mobile phone calls using native, mobile data, or WiFi networks, respectively
- $outputpower_{native}$, $outputpower_{data}$, $outputpower_{wifi}$ are the output powers in mW of the device (mobile phone) using native, mobile data, or WiFi networks, respectively

Value

Momentary SAR (mSAR) in mW/kg

See Also

mpc_pwr_native(), mpc_pwr_data(), mpc_pwr_wifi(), mpc_sar_native(), mpc_sar_data(), mpc_sar_wifi()

Examples

```
mpc_msar(
tissue          = "body",
prop_native     = 0.7,
prop_data       = 0.2,
prop_wifi       = 0.1,
headp_prop      = 0.17,
ear_prop        = 0.67,
speaker_prop    = 0.17,
urbanicity      = "suburban",
use_5g          = TRUE,
travel_time     = 1800)
```

---

mpc_pwr_data *Calculate Mobile Phone Output Power during Data Mobile Phone Calls*

Description

Calculates the mobile phone output power during calls using mobile data.

Usage

```
mpc_pwr_data(band, urbanicity, travel_time, params = load_params())
```

Arguments

band	Technology (3g, 4g, or 5g) used for the call
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends in the technology used and the location in which the call is performed (urbanicity, indoors/outdoors/commuting)

Value

Output power in mW

Examples

```
mpc_pwr_data(
band        = "4g",
urbanicity  = "suburban",
travel_time = 1800)
```

mpc_pwr_native *Calculate Mobile Phone Output Power during Native Mobile Phone Calls*

Description

Calculates the mobile phone output power during calls using native network.

Usage

```
mpc_pwr_native(band, urbanicity, travel_time, params = load_params())
```

Arguments

band	Technology (2G, 3g, 4g, or 5g) used for the call
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends in the technology used and the location in which the call is performed (urbanicity, indoors/outdoors/commuting)

Value

Output power in mW

Examples

```
mpc_pwr_native(
band        = "4g",
urbanicity  = "suburban",
travel_time = 1800)
```

mpc_pwr_wifi — *Calculate Mobile Phone Output Power during WiFi Mobile Phone Calls*

Description

Calculates the mobile phone output power during calls using WiFi connection.

Usage

```
mpc_pwr_wifi(band, params = load_params())
```

Arguments

`band`	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz) used for the call
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the frequency band (2.4 GHz or 5.0 GHz)

Value

Output power in mW

Examples

```
mpc_pwr_wifi(
band  = "5")
```

mpc_sar_data — *Calculate nSAR during Data Mobile Phone Calls*

Description

Calculates tissue-specific nSAR from mobile phone calls using mobile data.

Usage

```
mpc_sar_data(
  tissue,
  band,
  headp_prop,
  ear_prop,
  speaker_prop,
  params = load_params()
)
```

Arguments

`tissue`	Tissue for which to calculate nSAR (default: "brain" or "body")
`band`	Technology (2G, 3g, 4g, or 5g) used for the call
`headp_prop`	Proportion of call performed with active Bluetooth connection to headphones
`ear_prop`	Proportion of call performed with phone held against ear
`speaker_prop`	Proportion of call performed in speaker mode
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue, the technology (2G, 3g, 4g or 5g), and the location of the phone during the call (against ear, with Bluetooth headphones, in speaker mode)

Value

nSAR in W/kg/W

Examples

```
mpc_sar_data(
tissue       = "brain",
band         = "4g",
headp_prop   = 0.17,
ear_prop     = 0.67,
speaker_prop = 0.17)
```

`mpc_sar_native` *Calculate nSAR during Native Mobile Phone Calls*

Description

Calculates tissue-specific nSAR from mobile phone calls using native network.

Usage

```
mpc_sar_native(
  tissue,
  band,
  headp_prop,
  ear_prop,
  speaker_prop,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Technology (2G, 3g, 4g, or 5g) used for the call
headp_prop	Proportion of call performed with active Bluetooth connection to headphones
ear_prop	Proportion of call performed with phone held against ear
speaker_prop	Proportion of call performed in speaker mode
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue, the technology (2G, 3g, 4g or 5g), and the location of the phone during the call (against ear, with Bluetooth headphones, in speaker mode)

Value

nSAR in W/kg/W

Examples

```
mpc_sar_native(
tissue       = "brain",
band         = "4g",
headp_prop   = 0.17,
ear_prop     = 0.67,
speaker_prop = 0.17)
```

---

mpc_sar_wifi *Calculate nSAR during WiFi Mobile Phone Calls*

Description

Calculates tissue-specific nSAR from mobile phone calls using WiFi connection.

Usage

```
mpc_sar_wifi(
  tissue,
  band,
  headp_prop,
  ear_prop,
  speaker_prop,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz) used for the call
headp_prop	Proportion of call performed with active Bluetooth connection to headphones
ear_prop	Proportion of call performed with phone held against ear
speaker_prop	Proportion of call performed in speaker mode
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue, the technology (2.4gHz or 5.0GHz), and the location of the phone during the call (against ear, with Bluetooth headphones, in speaker mode)

Value

nSAR in W/kg/W

Examples

```
mpc_sar_wifi(
tissue       = "brain",
band         = "2",
headp_prop   = 0.17,
ear_prop     = 0.67,
speaker_prop = 0.17)
```

---

mpd_msar — *Calculation of mSAR from Mobile Phone Data*

Description

Calculates mSAR (momentary specifc absorption rate) from data (mobile data and WiFi) use for a specific tissue

Usage

```
mpd_msar(
  tissue,
  duration_low,
  duration_lowmed,
  duration_medhigh,
  duration_high,
  use_5g,
  wifi_prop_home,
  wifi_prop_work,
  wifi_prop_travel,
  urbanicity,
  travel_time,
  params = load_params()
)
```

Arguments

`tissue`	Tissue for which to calculate mSAR (default: "brain" or "body")
`duration_low`	Duration (in seconds per day) of low output power activities
`duration_lowmed`	Duration (in seconds per day) of low-medium output power activities
`duration_medhigh`	Duration (in seconds per day) of medium-high output power activities
`duration_high`	Duration (in seconds per day) of high output power activities
`use_5g`	TRUE if 5G services are used for data transfer, FALSE if not
`wifi_prop_home`	Proportion of time connected to WiFi (vs mobile data) at home
`wifi_prop_work`	Proportion of time connected to WiFi (vs mobile data) at school / work
`wifi_prop_travel`	Proportion of time connected to WiFi (vs mobile data) while commuting
`urbanicity`	Urbanicity of home / workplace (rural, suburban, or urban)
`travel_time`	Time spent commuting in seconds per day
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as

$$mSAR = nSAR_{mobiledata} \times outputpower_{mobiledata} + nSAR_{wifi} \times outputpower_{wifi}$$

where

- $nSAR_{data}$, $nSAR_{wifi}$ are the nSAR (normalized specific absorption rates, in W/kg/W) from data use using mobile data or WiFi networks, respectively
- $outputpower_{data}$, $outputpower_{wifi}$ are the output powers (mW) of the device (mobilephone) using mobile data or WiFi networks, respectively

Value

Momentary SAR (mSAR) from mobile data use in mW/kg

See Also

`mpd_pwr_data()`, `mpd_pwr_wifi()`, `mpd_sar_data()`, `mpd_sar_wifi()`

Examples

```
mpd_msar(
tissue          = "brain",
duration_low    = 4860,
duration_lowmed  = 540,
duration_medhigh = 4860,
duration_high    = 540,
use_5g          = FALSE,
wifi_prop_home  = 0.5,
wifi_prop_work  = 0.5,
```

```
wifi_prop_travel = 0.5,
urbanicity       = "suburban",
travel_time      = 1800)
```

---

mpd_pwr_data *Calculate Mobile Phone Output Power during Mobile Data Use*

Description

Calculates the mobile phone output power during mobile data use.

Usage

```
mpd_pwr_data(
  band,
  duration_low,
  duration_lowmed,
  duration_medhigh,
  duration_high,
  urbanicity,
  travel_time,
  params = load_params()
)
```

Arguments

band	Technology (3g, 4g, or 5g)
duration_low	Duration (in seconds per day) of low output power activities
duration_lowmed	Duration (in seconds per day) of low-medium output power activities
duration_medhigh	Duration (in seconds per day) of medium-high output power activities
duration_high	Duration (in seconds per day) of high output power activities
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends in the technology used, the location in which the mobile data network (urbanicity, indoors/outdoors/commuting), and the activity the phone is used for.

Value

Output power in mW

Examples

```
mpd_pwr_data(
band            = "4g",
duration_low    = 4860,
duration_lowmed  = 540,
duration_medhigh = 4860,
duration_high   = 540,
urbanicity      = "rural",
travel_time     = 1800)
```

---

mpd_pwr_wifi — *Calculate Mobile Phone Output Power during WiFi Use*

Description

Calculates the mobile phone output power during WiFi use

Usage

```
mpd_pwr_wifi(
  band,
  duration_low,
  duration_lowmed,
  duration_medhigh,
  duration_high,
  params = load_params()
)
```

Arguments

`band`	WiFi frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
`duration_low`	Duration (in seconds per day) of low output power activities
`duration_lowmed`	Duration (in seconds per day) of low-medium output power activities
`duration_medhigh`	Duration (in seconds per day) of medium-high output power activities
`duration_high`	Duration (in seconds per day) of high output power activities
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the frequency band (2.4 GHz or 5.0 GHz) and the type of activity.

Value

Output power in mW

Examples

```
mpd_pwr_wifi(
band            = "2",
duration_low    = 4860,
duration_lowmed  = 540,
duration_medhigh = 4860,
duration_high    = 540)
```

mpd_sar_data — *Calculate nSAR during mobile data use on mobile phone*

Description

Calculates tissue-specific nSAR from mobile data use on mobile phone.

Usage

```
mpd_sar_data(band, tissue, params = load_params())
```

Arguments

band	Technology ("3g", "4g", or "5g")
tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the frequency band.

Value

nSAR in W/kg/W

Examples

```
mpd_sar_data(
tissue       = "brain",
band         = "5g")
```

`mpd_sar_wifi` *Calculate nSAR during WiFi Use on mobile phone*

Description

Calculates tissue-specific nSAR from WiFi use on mobile phone.

Usage

```
mpd_sar_wifi(band, tissue, params = load_params())
```

Arguments

`band`	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
`tissue`	Tissue for which to calculate nSAR (default: "brain" or "body")
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the frequency band.

Value

nSAR in W/kg/W

Examples

```
mpd_sar_wifi(
tissue      = "brain",
band        = "2")
```

`other_dose_wrapper` *Calculate RF-EMF Dose from Other Sources*

Description

Wrapper function that calculates total dose from additional devices and activities, as used in the GOLIAT consortium study dose calculation.

Usage

```
other_dose_wrapper(
  tissue,
  duration_hotspot,
  duration_smartwatch,
  duration_vr,
  duration_headphones,
  duration_gaming,
  params = load_params()
)
```

Arguments

`tissue` Tissue for which to calculate dose (default: "brain" or "body")

`duration_hotspot` Duration of time hotspot was active in s/day

`duration_smartwatch` Duration of time smartwatch was used in s/day

`duration_vr` Duration of time VR headset was used in s/day

`duration_headphones` Duration of time Bluetooth headphones were used in s/day

`duration_gaming` Duration of gaming with portable console in s/day

`params` Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from other sources takes into account 5 additional sources:

1. Hotspot (smartphone)
2. Smart watch
3. Virtual reality (VR) headset
4. Bluetooth headphones
5. Portable gaming consoles

The dose is calculated as

$$Dose_{other} = Dose_{hotspot} + Dose_{smartwatch} + Dose_{vr} + Dose_{headphones} + Dose_{gaming}$$

The dose from each source is calculated from the corresponding dose functions.

Value

Tissue-specific RF-EMF dose from other sources in mJ/kg/day

See Also

hotspot_dose(), smartwatch_dose(), vr_dose(), headphones_dose(), gaming_dose()

Examples

```
other_dose_wrapper(
  tissue              = "brain",
  duration_hotspot    = 600,
  duration_smartwatch = 86400,
  duration_vr         = 0,
  duration_headphones = 8280,
  duration_gaming     = 3600)
```

smartwatch_dose *Calculate RF-EMF Dose from Smart Watch*

Description

Calculates the tissue-specific RF-EMF dose from smart-watch use

Usage

```
smartwatch_dose(tissue, duration_smartwatch, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_smartwatch	duration of time smartwatch was worn in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from smart-watch use is calculated as:

$$Dose_{smartwatch} = Dose_{watch} + Dose_{phone}$$

Where:

- $Dose_{watch}$ is the dose contribution from the smart-watch itself
- $Dose_{phone}$ is the dose contribution from the phone connection to the smart-watch

Value

Tissue-specific RF-EMF dose from smart-watch use in mJ/kg/day

See Also

smartwatch_watch_dose(), smartwatch_phone_dose()

Examples

```
smartwatch_dose(
  tissue              = "body",
  duration_smartwatch = 6000)
```

smartwatch_phone_dose *Calculate RF-EMF Dose from Smart Watch (phone contribution only)*

Description

Calculates the tissue-specific RF-EMF dose from smart-watch use (phone contribution only).

Usage

```
smartwatch_phone_dose(tissue, duration_smartwatch, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`duration_smartwatch`	duration of time smart-watch was worn in s/day
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from smart-watch use (phone contribution) is calculated as:

$$Dose_{phone} = mSAR_{phone} * duration_{phone}$$

Where:

- $mSAR_{phone}$ is the momentary SAR
- $duration_{phone}$ is the use duration

Value

Tissue-specific RF-EMF dose from smart-watch use (phone contribution only) in mJ/kg/day

Examples

```
smartwatch_phone_dose(
  tissue             = "body",
  duration_smartwatch = 6000)
```

smartwatch_watch_dose *Calculate RF-EMF Dose from Smart Watch (watch contribution only)*

Description

Calculates the tissue-specific RF-EMF dose from smart-watch use (watch contribution only).

Usage

```
smartwatch_watch_dose(tissue, duration_smartwatch, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_smartwatch	duration of time smart-watch was worn in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from smart-watch use (watch contribution) is calculated as:

$$Dose_{watch} = mSAR_{watch} * duration_{watch}$$

Where:

- $mSAR_{watch}$ is the momentary SAR
- $duration_{watch}$ is the use duration

Value

Tissue-specific RF-EMF dose from smart-watch use (watch contribution only) in mJ/kg/day

Examples

```
smartwatch_watch_dose(
  tissue              = "body",
  duration_smartwatch = 6000)
```

tablet_dose — *Calculate RF-EMF Dose from Tablet Use*

Description

Calculates the tissue-specific RF-EMF dose from tablet use.

Usage

```
tablet_dose(
  tissue,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

`tissue`	Tissue for which to calculate dose (default: "brain" or "body")
`dur_low`	Duration (in seconds per day) of low output power activities on tablet
`dur_lowmed`	Duration (in seconds per day) of low-medium output power activities on tablet
`dur_medhigh`	Duration (in seconds per day) of medium-high output power activities on tablet
`dur_high`	Duration (in seconds per day) of high output power activities on tablet
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The tablet RF-EMF dose is calculated as:

$$Dose_{tablet} = mSAR_{tablet} * duration_{tablet}$$

Where:

- $Dose_{tablet}$ is the dose from tablet use
- $mSAR_{tablet}$ is the momentary SAR value in mJ/kg
- $duration_{tablet}$ is the duration of tablet use

We distinguish between 4 types of activities:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific RF-EMF dose from tablet use in mJ/kg/day

See Also

tablet_msar()

Examples

```
tablet_dose(
  tissue      = "brain",
  dur_low     = 0,
  dur_lowmed  = 681,
  dur_medhigh = 0,
  dur_high    = 0)
```

---

tablet_msar — *Calculate mSAR from Tablet Use*

Description

Calculates the tissue-specific momentary SAR (mSAR) from tablet use.

Usage

```
tablet_msar(
  tissue,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
dur_low	Duration (in seconds per day) of low output power activities on tablet
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on tablet
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on tablet
dur_high	Duration (in seconds per day) of high output power activities on tablet
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as:

$$mSAR_{tablet} = nSAR_{tablet} * outputpower_{tablet}$$

Where:

- $mSAR_{tablet}$ is the mSAR from tablet use
- $nSAR_{tablet}$ is the normalised SAR value in mJ/kg
- $outputpower_{tablet}$ is the duration of tablet use

We distinguish between 4 types of activities:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific mSAR from tablet use in mW/kg

See Also

tablet_pwr(), tablet_sar()

Examples

```
tablet_msar(
  tissue      = "brain",
  dur_low     = 0,
  dur_lowmed  = 1230,
  dur_medhigh = 0,
  dur_high    = 0)
```

---

tablet_pwr *Calculate Tablet Output Power*

Description

Calculates the tablet output power during use

Usage

```
tablet_pwr(
  band,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

band	WiFi frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
dur_low	Duration (in seconds per day) of low output power activities on tablet
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on tablet
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on tablet
dur_high	Duration (in seconds per day) of high output power activities on tablet
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the frequency band (2.4 GHz or 5.0 GHz) and the type of activity.

Value

Output power in mW

Examples

```
tablet_pwr(
  band        = "5",
  dur_low     = 0,
  dur_lowmed  = 681,
  dur_medhigh = 0,
  dur_high    = 0)
```

---

tablet_sar *Calculate nSAR during Tablet Use*

Description

Calculates tissue-specific nSAR from tablet use.

Usage

```
tablet_sar(tissue, band, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the frequency band.

Value

nSAR in W/kg/W

Examples

```
tablet_sar(
tissue      = "brain",
band        = "2")
```

`total_dose` *Calculate Total RF-EMF Dose from All Sources*

Description

Calculates the total RF-EMF dose from all available RF-EMF sources for a specific tissue.

Usage

```
total_dose(sample, tissue, params = load_params())
```

Arguments

sample	A list with input values for a single observation. List elements must match the names and data types of the provided example dataset and must not be NA.
tissue	Tissue for which to calculate dose (default: "brain" or "body")
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

A detailed description of the dose calculation will follow.

Value

Total RF-EMF dose in mJ/kg/day for the specified tissue.

vr_dose *Calculate RF-EMF Dose from Virtual Reality Headset*

Description

Calculates the tissue-specific RF-EMF dose from VR headset

Usage

```
vr_dose(tissue, duration_vr, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_vr	duration of time VR headset was used in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from VR headset is calculated as:

$$Dose_{vr} = mSAR_{vr} * duration_{vr}$$

Where:

- $mSAR_{vr}$ is the mSAR of virtual reality headet use
- $Dduration_{vr}$ is the duration of VR headset use in s/day

Value

Tissue-specific RF-EMF dose from VR headset use in mJ/kg/day

Examples

```
vr_dose(
  tissue      = "brain",
  duration_vr = 600)
```

wifi_dose *Calculate RF-EMF Dose from WiFi Router*

Description

Calculates the tissue-specific RF-EMF dose from WiFi router

Usage

```
wifi_dose(tissue, travel_time, wifi_prop_travel, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate nSAR (default: "brain" or "body")
`travel_time`	Time spent commuting in seconds per day
`wifi_prop_travel`	Proportion of time connected to WiFi (vs mobile data) while commuting
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from WiFi router is calculated as:

$$Dose_{WiFi} = mSAR_{farfield} * duration_{WiFi}$$

Where:

- $Dose_{WiFi}$ is the dose from WiFi Router
- $mSAR_{WiFi}$ is the momentary SAR value in mJ/kg
- $duration_{WiFi}$ is the duration of time exposed to a WiFi router in seconds/day

Value

Tissue-specific RF-EMF dose from Wifi router in mJ/kg/day

See Also

`wifi_msar()`

Examples

```
wifi_dose(
  tissue           = "body",
  travel_time      = 1800,
  wifi_prop_travel = 0.5)
```

`wifi_msar` *Calculate mSAR of WiFi Router*

Description

Calculates the mSAR of WiFi Router

Usage

```
wifi_msar(tissue, travel_time, wifi_prop_travel, params = load_params())
```

Arguments

`tissue`	Tissue for which to calculate nSAR (default: "brain" or "body")
`travel_time`	Time spent commuting in seconds per day
`wifi_prop_travel`	Proportion of time connected to WiFi (vs mobile data) while commuting
`params`	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the time spent commuting and the proportion of time a WiFi connection (vs mobile data) is used while commuting.

Value

mSAR in W/kg/W/m**2

See Also

`wifi_pwr()`, `wifi_sar()`

Examples

```
wifi_msar(
  tissue           = "body",
  travel_time      = 1800,
  wifi_prop_travel = 0.5)
```

wifi_pwr — *Calculate Output Power of WiFi Router*

Description

Calculates the output power of WiFi router

Usage

```
wifi_pwr(band, params = load_params())
```

Arguments

band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the WiFi frequency band.

Value

Output power in mW/m**2

Examples

```
wifi_pwr(
band = "2")
```

wifi_sar — *Calculate nSAR from WiFi Router*

Description

Calculates tissue-specific nSAR from WiFi Router

Usage

```
wifi_sar(tissue, band, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the WiFi frequency band.

Value

nSAR in W/kg/W/m**2

Examples

```
wifi_sar(
tissue = "brain",
band   = "5")
```

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